LEVEL UP ACADEMY

Class 9 | CBSE / GSEB | Science

Unit Test: Sound

Chapter / Topic: Sound (topic-based)

Name: _____ Roll No.: _____ Date: _____

Time: 50 minutes **Max. Marks: 25**

General instructions:

1. All questions are compulsory. Internal choice is given in some questions.
2. Section A: 1 mark each. Section B: 2 marks each. Section C: 3 marks each. Section D: 5 marks. Section E: case-based, 4 marks.
3. Show the formula, substitution and unit in every numerical.

SECTION A

6 marks

Choose the correct option.

Q1. Sound travels fastest in: **[1]**

(a) air (b) water
(c) steel (d) a vacuum

Q2. The SI unit of frequency is the: **[1]**

(a) metre (b) second
(c) decibel (d) hertz

Q3. Sound waves in air are: **[1]**

(a) transverse (b) longitudinal
(c) electromagnetic (d) not waves at all

Q4. Sound of frequency higher than 20 kHz is called: **[1]**

(a) infrasonic (b) ultrasonic
(c) audible (d) supersonic

Q5. A sound has a frequency of 500 Hz. Its time period is: **[1]**

(a) 0.002 s (b) 0.02 s
(c) 2 s (d) 500 s

Q6. **Assertion (A):** We see a lightning flash before we hear the thunder.
Reason (R): Light travels much faster than sound. **[1]**

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true, but R is false.
(d) A is false, but R is true.

SECTION B

4 marks

Q7. What are compressions and rarefactions? **[2]**

OR

What is reverberation? How can it be reduced in a big hall?

Q8. Sound travels at 340 m/s in air. Find the frequency of a sound wave of wavelength 1.7 m. **[2]**

SECTION C

6 marks

Q9. A boy claps his hands in front of a tall building and hears the echo 3 s later. (Speed of sound = 340 m/s) **[3]**

(i) What total distance has the sound travelled? **[1]**

(ii) How far is the building from the boy? **[1]**

(iii) What is the minimum distance needed to hear a distinct echo at this speed of sound? **[1]**

Q10. Give three uses of ultrasound. **[3]**

OR

What is SONAR? Explain how it is used to find the depth of the sea.

SECTION D

5 marks

Q11. Answer the following: **[5]**

(i) Explain how sound travels through air from a vibrating object to our ears. **[3]**

(ii) A ship sends out an ultrasonic pulse and receives the echo from the sea bed after 4 s. The speed of sound in sea water is 1500 m/s. How deep is the sea? **[2]**

SECTION E

4 marks

Read the passage and answer the questions that follow.

Q12. The outer ear collects sound and sends it through the ear canal to the eardrum, which starts vibrating. In the middle ear, three tiny bones (the hammer, anvil and stirrup) increase these vibrations several times and pass them to the inner ear. In the inner ear, the cochlea converts the pressure variations into electrical signals, which the auditory nerve carries to the brain. **[4]**

(i) Which part of the ear vibrates first when sound reaches it? **[1]**

(ii) Name the three bones of the middle ear. **[1]**

(iii) What is the role of the cochlea? Why can very loud sounds be harmful? **[2]**

~ All the best! ~